Ziphozihle Maqopi

66/69

95%

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ROTATING ELECTRICAL MACHINERY FORMATIVE ASSESSMENT

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About the Formative Assessment	
Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

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Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Ziphozinle Mqopi	Manager:
Trainer:	Mentor / Coach:
Facilitator: Balennah Yabadi	
Assessor: Balennah Yabadi	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Ziphozinle Mqopi

Learner's signature: _____ Date: _____

Facilitator Name: Balennah Yabadi

Facilitator's signature: _____ Date: _____

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Ziphozihle Maopi	Signature
Facilitator Name:	Signature:

Forr

LEARNING UNIT 1:
KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction b) Describe with the aid of diagrams, the configuration of motor connections c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	5 5 10

1a) Induction Motor:

Stator: Contains stator windings that produce the rotating magnetic field. These windings are housed in slots in a laminated steel core

Rotor: Two main types: Squirrel Cage: Consists of copper or aluminium bars embedded in slots in a laminated steel core. These bars are short-circuited at both ends by end rings. ~~The second is a Wound rotor. A wound rotor~~ Wound rotor: has a wound rotor with insulated windings connected to slip rings

1a) Three-phase squirrel cage Induction Motor

Construction: Stator → Laminated steel core with slots. Carries a 3-phase distributed winding connected to AC supply. Produces a rotating magnetic field

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LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of alternating current motors	4
	b) Describe and explain how tests are conducted on alternating current motors	4
	c) Describe and explain protection devices used for alternating current motors	5
	d) Compare the advantages and disadvantages of single- and three-phase motors	6

2a) The operation of alternating current motors is based on the prinaple of electromagnetic induction. When alternating current (AC) flows through the stator windings, it creates a rotating magnetic field. This magnetic field interacts with the rotor, which is usually made of conductive material and is free to rotate. The interation between the stator's rotating magnetic field and the rotor's magnetic induces a torque, causing the rotor to turn and drive mechanical systems. This prinaple helps to design and fuction AC motors

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LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201	Construction of direct current motors
Learning Outcome 2:	KT0202	Principles of operation for direct current motors
Learning Outcome 3:	KT0203	Configuration of direct current motor connections
Learning Outcome 4:	KT0204	Types and application of direct current motors
Learning Outcome 5:	KT0205	Testing principles of direct current motors
Learning Outcome 6:	KT0206	Protection of direct current motors
Learning Outcome 7:	KT0207	Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) List the different types of direct current motors and describe their construction	2
	b) Describe the applications of different types of direct current motors	2
	c) Describe with the aid of diagrams, the configuration of motor connections, power circuits and control circuits.	2
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation, short circuits.	2
	e) Calculate direct current motor properties	2

2a)

- DC Shunt Motor

Construction: The Field winding is connected in parallel with an armature. The field winding has many turns of thin wire, giving it high resistance. The armature and field both receive full supply voltage. It has a commutator and carbon brushes to feed DC to rotating armature

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Power Circuit



d.) Time Delay: Overload protection devices with time delay features are designed to allow temporary overloads to pass without tripping the circuit immediately. This is beneficial in scenarios like motor start-up or temporary high current demands. Where a brief overload is expected. The delay ensures that the protection device does not trip unnecessarily preventing frequent interruptions due to transient condition.

Current Rating: The current rating of overload protection devices determines the maximum current threshold at which the device will trip to protect the motor. It is crucial to select a protection device with a current rating suitable for the motor's normal operating current. If the current is too low, the protection device may trip frequently under normal operating conditions.

e) Back emf: $E_b = V - I_a R_a$

Power developed: $P_a = E_b I_a$

where $I_a = I_L - I_{sh}$ and $I_{sh} = \frac{V}{R_{sh}}$

Working:

$I_{sh} = \frac{220}{110} = 2A$

$I_a = 50 - 2 = 48A$

$E_b = 220 - 48 \times 0,2 = 210,4V$

$P_a = 210,4 \times 48 = 10,1 KW$

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coil. A commutator and brushes mechanically rectify this AC into pulsating DC at the output terminals. In contrast, an ~~alterno~~ alternator typically has a rotating DC-excited field and a stationary armature. As the rotor turns, its magnetic field cuts the stator conductors, inducing alternating current directly in the stationary windings. This alternating current is collected via slip rings with no mechanical rectification. Thus while, both machines convert mechanical energy to electrical energy using the same induction principle.

b)

1) DC shunt ~~Motor~~ Generator: Gives nearly a constant voltage. It is used for battery charging, electroplating and as exciters for alternators.

2) DC Series Generator: Voltage increase with load. Used as voltage boosters in DC feeders and for arc welding

3) Synchronous Alternator: Main machine in power stations. Used in thermal, hydro, and nuclear plants to generate AC for the national grid

4) Automotive Alternator: Small 3-phase AC generator with rectifiers. Used in vechicles to charge the battery and supply all ~~eet~~ electrical loads.

~~c) Overcurrent Prot~~

c) Overcurrent Protection: Uses overcurrent relays and circuit breakers. Trips the machine if the load current exceeds rating, preventing winding overheating and insulation damage

Differential Protection: Compares current entering and leaving the stator winding using CTs. If the difference occurs due to internal fault, the ~~re~~ relay trips instantly. Protects against phase-to-phase and phase-to-earth faults.

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SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	KT0301 Construction of alternators and generators			✓
2	KT0302 Principle of operation of alternators and generators			✓
3	KT0303 Configuration of alternator and generator connections			✓
4	KT0304 Types and application of alternators and generators			✓
5	KT0305 Testing principles of alternators and generators			✓
6	KT0306 Protection of alternators and generators			✓
7	KT0307 Calculate properties of alternators and generators			✓

Learner Signature **Date**

Facilitators' Signature **Date**

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DECLARATION BY LEARNER			
I, Ziphozinle Mqopi .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I Baienah Tjabadi ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

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Notification of Assessment

Assessment will take place as follows:

	Theory	**Practical**
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

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5. Complete learner toolkit

6. Complete assessor toolkit